

Draft - expert review pending

This unit is complete and passes the platform's automated checks, but no reviewer has read it yet. Treat its sources and claims with care.

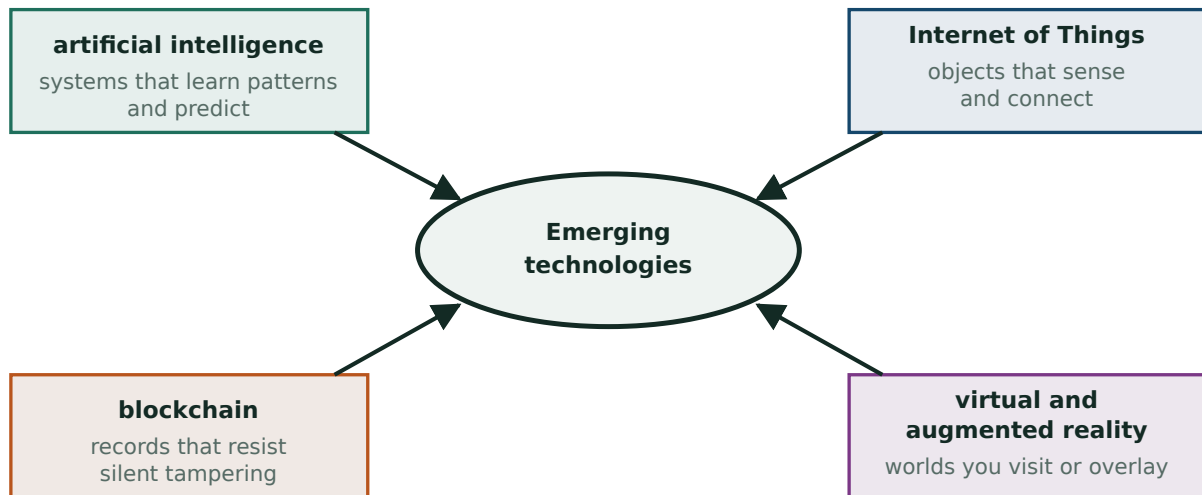
Topic 6.3 - Artificial Intelligence and the Internet of Things

A real classroom situation

Two scenes from one week at a Sukkur school. In the first, a trainee teacher spends an evening with an AI chatbot, asking it to draft a Class 6 quiz on photosynthesis; in ten minutes she has twenty questions - and, on checking, finds two facts quietly wrong. In the second, the head teacher's office gets a smart attendance device that logs each teacher's entry with a fingerprint and posts the day's record to the office system automatically. Nobody typed anything. One technology learned her subject; the other sensed her school. Both arrived without instructions about trust, privacy or error - and that is what this topic supplies.

The emerging-technology family

Same closing question for all four: what does it do for this lesson, at this cost, with these cautions?



Each moves a task from people to systems - and none of them moves responsibility.

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Explanation

Artificial Intelligence

Artificial intelligence (AI) is the field of building systems that perform tasks which normally need human intelligence - recognising speech and images, translating languages, answering questions, drafting text. Modern AI mostly works by learning patterns from very large amounts of example data: show a system enough labelled photographs and it learns to label new ones; feed it enough text and it learns to produce fluent sentences. It does not understand as a person does; it predicts. That single fact explains both its power (fluency, speed, tirelessness) and its known failure (it can be confidently, fluently wrong - the two wrong facts in the trainee's quiz).

For a teacher, AI's present uses are already practical: drafting quizzes, worksheets and letters to review and correct; translation support for bilingual material; speech-to-text for accessibility; and adaptive learning apps that adjust difficulty to the learner. UNESCO's guidance for policy-makers frames the education questions precisely: AI in education should enhance human capacity rather than replace teachers, should protect learners' data and agency, and

should be governed by human accountability for its outcomes (UNESCO, 2021). The trainee's rule of thumb follows: AI drafts, the teacher decides. Nothing goes to a pupil unreviewed.

Internet of Things

The Internet of Things (IoT) is the family of ordinary objects given sensors, small processors and a network connection, so they can measure and report without a human typing anything: the biometric attendance device, a smart electricity meter, a water-level sensor in a tank, a vehicle tracker. Bourgeois and colleagues (2019) define it as devices "embedded into a variety of objects" and connected online, and offer the memorable test: think of IoT as devices you would not normally consider being connected to the internet. The ITU's standard definition makes the same point in formal terms, calling it the network of connected things with sensing capability (ITU-T, 2012).

AI and IoT in education: uses and cautions

Technology	One school use	The caution
AI drafting tools	quiz and worksheet drafts in minutes	fluent errors: a teacher must review every fact
AI translation	bilingual material support	nuance can drift: check key terms
IoT attendance device	automatic staff and pupil records	biometric data is sensitive: strictest handling
IoT water or power sensor	tank levels and bills without site visits	sensors fail silently: keep a manual check
UNESCO's rule for AI in education: enhance human capacity rather than replace it, with human accountability for outcomes.		

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In a Pakistani school, IoT's honest use-cases are the unglamorous ones: attendance, utilities and security, where automatic records beat manual registers. The cautions are Unit 4's, sharpened. A device that senses automatically also collects automatically, and biometric data is sensitive personal data under any standard - the four questions (why, where, who, how

long) apply to a fingerprint reader exactly as they applied to the app vendor. And IoT devices are famous for forgotten passwords: the device that ships with a default password and keeps it is an open door on the school's network.

Both technologies share the deeper pattern of this unit: each moves a task from people to systems, and each therefore moves responsibility too. The teacher who lets AI draft still owns the facts; the school that installs sensors still owns the data. Emergence never transfers accountability.

Activity: The briefing notebook

Work in groups of three. About 20 minutes.

1. Each group takes one technology from this topic: AI drafting, AI translation, IoT attendance, or IoT utilities.
2. For your technology, write the briefing skeleton a head teacher would need: what it is (one sentence), one concrete school use, one caution with its Unit 4 or Unit 5 connection, and one question to ask a vendor.
3. Test the caution: write the one sentence you would tell a colleague who is about to adopt the technology without thinking.
4. Groups present skeletons in two minutes each; the class assembles the beginnings of the emerging-technology briefing for the unit's practicum task.

Check your understanding

1. Define artificial intelligence and the Internet of Things in one sentence each.
(Remember)
2. Explain why AI can be "confidently, fluently wrong", using how it works.
(Understand)
3. A school installs a biometric attendance system. Name the data it collects and the four questions the school must answer before and during its use.
(Apply)
4. State UNESCO's framing rule for AI in education in your own words.
(Understand)

Summary

Artificial intelligence learns patterns from vast examples and predicts - drafting, translating, recognising - with fluency that can hide error, so AI drafts and the teacher decides, under UNESCO's rule of enhancing rather than replacing human capacity. The Internet of Things embeds sensing and connection in ordinary objects - attendance, meters, tanks - automating records that once needed typing, at the price of automatic collection of (often sensitive) data and forgotten default passwords. Both move tasks to systems; neither moves responsibility.

Self-assessment checklist

Sign in to sync your ticks across devices.

- ☐ I can define AI and IoT and give one school use of each.
- ☐ I can explain the fluent-error problem and the review rule it demands.
- ☐ I can apply the four data questions to a biometric device.

Try this at your practicum school

Find one automatic or smart device in the school (attendance machine, security camera, UPS, meter). Write its briefing skeleton: what it senses, where the data goes, who can see it, and the one question you would ask about it.

Summative task

Write a one-page emerging-technology briefing for your head teacher on EITHER an AI drafting tool OR a biometric attendance system: what it is, what it would do for the school, the cautions (facts, data, accountability), and your recommendation with conditions. Ground every caution in a rule from Units 4 or 5.

Marking guide (10 marks):

- The technology explained accurately in plain language: 2 marks.

- The school use is concrete and realistic for the school's resources: 2 marks.
- The cautions are tied to named rules (fluent errors, sensitive data, four questions) (Analyse): 4 marks.
- The recommendation carries conditions, not just yes or no: 2 marks.

Further reading

- UNESCO. (2021). *AI and education: guidance for policy-makers*. <https://unesdoc.unesco.org/ark:/48223/pf0000376709>
- Bourgeois, D. T., Smith, J. L., Wang, S., & Mortati, J. (2019). *Information Systems for Business and Beyond* (2nd ed.), chapter 13. Open Textbook Library. <https://open.umn.edu/opentextbooks/textbooks/information-systems-for-business-and-beyond>
- ITU-T. (2012). *Recommendation ITU-T Y.2060: Overview of the Internet of things*. <https://www.itu.int/rec/T-REC-Y.2060-201206-I>

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